Study Overview

Brief Summary

ZEPHIR-02 is a multicentre, open-label phase II study that will enroll subjects with HER2-positive advanced/metastatic breast cancer (mBC) who have experienced disease progression under trastuzumab deruxtecan (T-DXd) in the metastatic setting.

All subjects will undergo baseline biopsy, blood collection, FDG-PET/CT and 89Zr-trastuzumab PET/CT (HER2-PET/CT) and will be classified as HER2-PET/CT positive or negative, as previously described in the ZEPHIR trial. Focusing on a central visual "patient-based" classification that captures the entire disease burden, a side-by-side display will be used, comparing baseline FDG-PET/CT (which identifies all FDG-positive metastases regardless of their HER2-imaging status) and HER2-PET/CT. Subjects will be categorized into two HER2-PET/CT patterns (positive vs. negative) based on proportion of FDG-avid tumor load with significant 89Zr-trastuzumab uptake.

Subjects classified as "positive" will receive T-DM1 as monotherapy, IV 3.6mg/kg every 3 weeks (21 days +- 3 days) until disease progression, unacceptable toxicity or request of the subject to withdraw from the study. FDG-PET/CT will be performed before cycle 2 of T-DM1 will serve as a research tool to correlate metabolic changes with clinical outcomes. Other FDG-PET/CT will be performed before cycle 4 of T-DM1 for assessment of response. Subjects who demonstrate a partial or complete response (responders) will continue treatment with T-DM1. Subjects who exhibit stable disease or disease progression (non-responders) will discontinue study treatment and enter the survival follow-up period. For responders, subsequent metabolic evaluations will be performed every 3 months, with FDG-PET/CT. Treatment response will be assessed according to metabolic response. For these subjects, mandatory blood samples will be obtained at all metabolic reassessments. Subjects with HER2-PET/CT classified as "negative" will receive treatment of physician's choice (TPC) as per the best local clinical practice and be out of the study.

All enrolled subjects will undergo a mandatory biopsy during the pre-treatment period.

The study also includes mandatory translational procedures (i.e. collection of tumour biopsy during pre-treatment period and blood samples at pre-specified time points) for exploratory molecular analyses.

Detailed Description

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All subjects will undergo baseline biopsy, blood collection, FDG-PET/CT and 89Zr-trastuzumab PET/CT (HER2-PET/CT) and will be classified as HER2-PET/CT positive or negative, as previously described in the ZEPHIR trial. Focusing on a central visual "patient-based" classification that captures the entire disease burden, a side-by-side display will be used, comparing baseline FDG-PET/CT (which identifies all FDG-positive metastases regardless of their HER2-imaging status) and HER2-PET/CT. Subjects will be categorized into two HER2-PET/CT patterns (positive vs. negative) based on proportion of FDG-avid tumour load with significant 89Zr-trastuzumab uptake. Lesion uptake will be considered significant/pertinent if it is visually higher than the local background.

HER2-PET/CT positive pattern: The entire or majority of the tumour load shows significant tracer uptake.
HER2-PET/CT negative pattern: The dominant part or all of the tumour load lacks significant tracer uptake.
Subjects classified as "positive" will receive T-DM1 as monotherapy, IV 3.6mg/kg every 3 weeks (21 days +- 3 days) until disease progression, unacceptable toxicity or request of the subject to withdraw from the study. FDG-PET/CT will

be performed before cycle 2 of T-DM1 for early assessment of response and then again before cycle 4. At the early FDG-PET/CT assessment (before cycle 2), response assessment will be done using a cut-off of 15% based on the CONSIST criteria. Subjects who demonstrate a partial or complete response (responders) and will continue treatment with T-DM1. Subjects who exhibit stable disease or disease progression (non-responders) will discontinue study treatment and enter the survival follow-up period. For responders, subsequent metabolic evaluations will be performed every 3 months, with FDG-PET/CT. Treatment response will be assessed according to metabolic response. For these subjects, blood samples will be obtained at all metabolic reassessments (mandatory).

The subjects with HER2 PET/CT classified as "negative" will receive treatment of physician's choice (TPC) as per the best local clinical practice. Subsequent treatment will be collected and the subject will enter survival follow-up.

All enrolled subjects will undergo a mandatory biopsy during the pre-treatment period. Of note, results of DNA sequencing on the biopsy will be communicated to the treating oncologist, to potentially inform post progression therapy.

Official Title
HER2 Molecular Imaging With 89Zr-trastuzumab PET/CT as a Predictive Biomarker for Antibody-drug Conjugate Sequencing in Patients With Advanced HER2-positive Breast Cancer

ontacts and Locations
This section provides contact details for people who can answer questions about joining this study, and information on where this study is taking place.

To learn more, please see the Contacts and Locations section in How to Read a Study Record.

Study Contact
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Participation Criteria
Researchers look for people who fit a certain description, called eligibility criteria. Some examples of these criteria are a person's general health condition or prior treatments.

For general information about clinical research, read Learn About Studies.
Eligibility Criteria
Description
Inclusion Criteria:

ECOG performance status â‰¤ 1
Must have histologically or cytologically confirmed progressive advanced/metastatic HER2-positive breast carcinoma as per the updated American Society of Clinical Oncology (ASCO) - College of American Pathologists (CAP) guidelines according to local testing. HER2 status may be determined in the primary breast cancer tumour or, when not available, in a metastatic lesion. Multifocal unilateral or bilateral breast adenocarcinoma tumours are allowed if all tested HER2-positive, according to local testing
Prior treatment with taxane, trastuzumab and pertuzumab (early or advanced setting) and T-DXd (metastatic setting). In order to be eligible, patients subjects must have received T-DXd as the last systemic metastatic treatment line before inclusion, and presented disease progression on this drug.
Prior therapy with tucatinib, trastuzumab, and capecitabine, in advanced setting, is permissible, provided that T-DXd serves as the last systemic metastatic treatment line before inclusion, and patient subject presented disease progression on this drug.

Life expectancy â‰¥ 6 months.
At screening FDG-PET at least two "target" lesions are required to fulfil the following criteria: (1) anatomically transaxial diameter â‰¥ 1.5 cm and (2) metabolically assessable with a maximum standard uptake value corrected for lean body mass (SUVmax) â‰¥ 1.5 x SUVmean + 2 standard deviations (SD) of the liver measured in a 3-cm-diameter spherical volume of interest (VOI) in normal liver parenchyma.
In case of suspected liver metastasis, a lesion should have a SUVmax â‰¥ 2 x SUVmean + 3 SD of the blood pool measured in a 1 cm-diameter VOI within descending thoracic aorta. Lesions pre-treated with irradiation are not eligible for consideration as "target" lesions.

Adequate Bone Marrow Function including:

Absolute Neutrophil Count (ANC) â‰¥1000/Î¼L or â‰¥1x109/L.
Platelets â‰¥100,000/Î¼L or â‰¥ 100 x 109/L.
Haemoglobin â‰¥ 9 g/dL.
Adequate Renal Function including serum creatinine â‰¤ 1.5 x upper limit of normal (ULN) or estimated creatinine clearance â‰¥ 60 ml/min as calculated using the method standard for the institution.
Adequate Liver Function, including all the following parameters:

Total serum bilirubin â‰¤ 1.5 x ULN unless the patient subject has documented Gilbert syndrome.
Aspartate and Alanine Aminotransferase (AST and ALT) â‰¤ 2.5x ULN.
Current left ventricular ejection fraction (LVEF) â‰¥ 50% on echocardiography or multiple-gated acquisition scanning and no history of a LVEF < 40% or symptomatic heart failure or a recent myocardial infarction.
Willingness to provide tumour tissue (mandatory biopsy) and blood samples (mandatory) for translational research activities.
Willingness to comply with the protocol for the duration of the study including treatment and scheduled visits and examinations.
Signed Informed Consent form (ICF) obtained prior to any study related procedure.
Inclusion criterion applicable to FRANCE only:

Affiliated to the French Social Security System
Exclusion Criteria:

Prior exposure to T-DM1 for the treatment of metastatic BC. For subjects exposed to T-DM1 for the treatment of early BC, subjects must not have relapsed while on or within 12 months of finishing treatment with T-DM1.
Brain metastasis as sole metastasis and/or symptomatic or requiring therapy to control symptoms.
History of interstitial lung disease / pneumonitis (grade 3 or 4) during the prior treatment with T-DXd.
Cardiopulmonary dysfunction as defined by any of the following:

Significant symptoms (Grade â‰¥ 2) relating to LV dysfunction, cardiac arrhythmia, or cardiac ischemia while or since receiving preoperative therapy.
Uncontrolled hypertension (systolic blood pressure > 180 mmHg and/or diastolic blood pressure > 100 mmHg)
Inadequately controlled angina, serious cardiac arrhythmia not controlled by adequate medication, severe conduction abnormality, or clinically significant valvular disease
Screening LVEF < 50% by either ECHO or MUGA
History of NCI CTCAE (Version 4.0) Grade â‰¥ 3 symptomatic congestive heart failure (CHF) or New York Heart Association (NYHA) criteria Class â‰¥ II
History of a decrease in LVEF to < 40% or symptomatic CHF with prior trastuzumab treatment (e.g., during preoperative therapy)
Myocardial infarction within 12 months prior to randomization
Requirement for continuous oxygen therapy
Known prior severe hypersensitivity to investigational product or any component in its formulations, including known severe hypersensitivity reactions to trastuzumab or excipients.
Contra-indication for treatment with T-DM1.
The number of subjects included in this trial, considered as "rapid progressors" (Rapid progressors defined as progressive disease within the first 6 months of T-DXd therapy) will be capped at 10% at enrolment (no more than 7 subjects out the 78 subjects planned to be recruited). After the first 7 "rapid progressors" included, progression within the first 6 months of T-DXd therapy will be considered as an exclusion criterion.
Any known liver disease, including known carriers of hepatitis B virus, hepatitis C, autoimmune hepatic disorders and sclerosing cholangitis.
Concurrent, serious, uncontrolled infections or known infection with HIV. Prior history of other invasive cancer in the past 5 years except basal or squamous cell carcinoma of skin that has been definitively treated.
Pregnant and/or lactating women, or intending to become pregnant during the study. Serum pregnancy test (for subjects of childbearing potential) positive within 15 days prior to enrolment.
Women of childbearing potential refusing to use one highly effective method of contraception from ICF signature, during the course of the study and at least 7 months after the last administration of T-DM1.
Men with childbearing potential partner refusing to use condom during the course of this study and for at least 7 months after the last administration of T-DM1.
Subject with a significant medical, neuro-psychiatric, or surgical condition, currently uncontrolled by treatment, which, in the principal investigator's opinion, may interfere with completion of the study.
Exclusion criterion applicable to FRANCE only:

Vulnerable persons according to the article L.1121-6 of the Public Health Code, adults who are the subject of a measure of legal protection or unable to express their consent according to article L.1121-8 of the Public Health Code.

Study Plan
This section provides details of the study plan, including how the study is designed and what the study is measuring.

Expand all / Collapse all
How is the study designed?
Design Details
Primary Purpose : Treatment
Allocation : Non-Randomized
Interventional Model : Sequential Assignment
Masking : None (Open Label)
Arms and Interventions

Participant Group/Arm Intervention/Treatment
Experimental: HER2 PET/CT "positive"
Subjects classified as HER2-PET/CT positive will receive T-DM1, IV 3.6mg/kg every 3 weeks, as monotherapy. HER2-PET/CT positive pattern: The entire or

majority of the tumour load shows significant tracer uptake.
Drug: Trastuzumab emtansine
T-DM1 will be administered IV at a dose of 3.6 mg/kg every 3 weeks. (21 days +/- 3 days) until disease progression, unacceptable toxicity or request of the subject to withdraw from the study. The total dose will depend on the subject's weight on day 1 of each T-DM1 cycle.

Other Names:
T-DM1
No Intervention: HER2 PET/CT "negative"
Subjects classified as HER2 PET/CT negative will receive treatment of physician's choice (TPC) as per the best local clinical practice. Subsequent treatment will be collected and the subject will enter survival follow-up. HER2-PET/CT negative pattern: The dominant part or all of the tumour load lacks significant tracer uptake.
What is the study measuring?
Primary Outcome Measures
Outcome Measure Measure Description Time Frame
Characterization of genomic alterations and HER2 expression. To characterize the tumour landscape (molecular alterations on tissue and HER2 expression and its heterogeneity) in HER2-positive mBC after at least 1 line of systemic therapy for advanced disease in order to better inform the oncologist and the subject regarding next treatment options. Through study completion, up to 2 years
Time to treatment failure (TTF)
The time to treatment failure (TTF), defined as the time from T-DM1 start to discontinuation for any reason, including disease progression (clinical or image-based on 18FDG-PET/CT), treatment toxicity or death in participants classified as "positive" (per metabolic response).

Time to Treatment Failure (TTF) of subjects classified as "positive" and its survival probability measures (median survival, survival rate at month 6) will be estimated using Kaplan-Meier method

During study treatment period (from C1 of T-DM1 until discontinuation), up to 1 year on average
Secondary Outcome Measures
Outcome Measure Measure Description Time Frame
Overall Survival under T-DM1 The Overall survival (OS) defined from the start of treatment to the date of death from any cause (subjects classified as "positive") Overall survival (OS) of subjects classified as "positive" will be analysed as time-to-event study and estimated using Kaplan-Meier method
From study treatment period until death, up to 1 year on average
Duration of Response under T-DM1 The Duration of response (DoR) defined as the time from the date of first documentation of response (CR or PR) to disease progression or death, in participants who achieve complete or partial response assessed by metabolic response (participants classified as "positive") DoR of subjects classified as "positive" will also be analysed as time-to-event study and estimated using Kaplan-Meier method During study treatment period until disease progression or death, up to 1 year on average
Disease Control Rate under T-DM1 The disease control rate (DCR) defined as absence of disease progression The DCR of subjects classified as "positive" will be described as proportion with two-sided 95% binomial proportion confidence intervals. From study treatment period until disease progression or death, up to 1 year on average
Safety of T-DM1 Incidence, nature, and severity of adverse events, including but not limited to drug induced liver injury The intensity of all AEs will be graded according to the NCI-CTCAE version 5.0 on a five-point scale (Grade 1 to 5) From study treatment period until progression disease, start of new anti-cancer treatment or death, up to 1 year on average
Other Outcome Measures
Outcome Measure Measure Description Time Frame
Correlation between "HER2 positivity" assessments in tumour tissue, HER2-PET/CT and ctDNA To characterize the correlation between different methods of HER2-

expression assessment (i.e., tissue evaluation [immunohistochemistry and in situ hybridization], molecular imaging [HER2-PET/CT], and circulating tumour DNA [ctDNA]) Through study completion, up to 2 years
Correlation between genomic and transcriptomic alterations with best overall response Through study completion, up to 2 years

Collaborators and Investigators
This is where you will find people and organizations involved with this study.
Sponsor
Jules Bordet Institute
Collaborators
Hoffmann-La Roche
Investigators
Study Chair:Martine Piccart, Md, PhD,Jules Bordet Institute

Study Record Dates
These dates track the progress of study record and summary results submissions to ClinicalTrials.gov. Study records and reported results are reviewed by the National Library of Medicine (NLM) to make sure they meet specific quality control standards before being posted on the public website.
Study Registration Dates
First Submitted
2024-09-04
First Submitted that Met QC Criteria
2024-09-11
First Posted
2024-09-19
Study Record Updates
Last Update Submitted that Met QC Criteria
2026-02-16
Last Update Posted
2026-02-19
Last Verified
2026-02